
WBI12 — Rapid Revision

IAL Biology Unit 2: Cells, Development, Biodiversity and Conservation

Last-minute cram sheets — designed to be scanned, not read

Topic priority ordered by frequency across 21 past papers (May 2019 – May 2026)

80 marks · 90 minutes · usually 8 questions

Q1–Q3 ≈ 21 marks | Q5–Q8 ≈ 49 marks

TRIAGE — what to review, in what order

Time left	Cover these	Why
30 min	Sheets 1, 3, 4, 5 — organelles, microscopy, gametes, mitosis	In 19–21 of 21 papers. Highest marks per minute.
2 hours	Add sheets 2, 8, 10 — prokaryotes + secretion, biodiversity index, plant cells and cellulose	In 16–19 of 21. “Cellulose” was the most frequent term in the mark schemes (76 mentions).
Half a day	Add sheets 6, 7, 9, 11, 12	9–14 of 21, but crossing over alone appeared 34 times in the mark schemes.
Always	Sheet 13 (calculations) and sheet 14 (phrase triggers)	Calculations in every paper; magnification is the commonest single question type.

TWO THINGS SPECIFIC TO THIS UNIT

1. Evolution is barely tested here. Across 21 papers: "natural selection" in 2, "speciation" in 1, "evolution" in 0. Adaptation and niche in 14–17. Don't spend your last hours on selection theory.

2. This unit rewards recall. "Give / state / name" = 9.1 marks per paper. The organelle table, gamete adaptations and mineral roles are marks you either have or don't.

SHEET 1 — CELL ULTRASTRUCTURE (21/21 papers)

Organelle	Structure	Function
Nucleus	Double membrane with pores ; chromatin; nucleolus	Holds DNA; transcription. Nucleolus makes ribosomes
Mitochondrion	Double membrane, inner folded into cristae ; matrix	Aerobic respiration → ATP . Cristae = large surface area
Rough ER	Membranes studded with ribosomes	Protein synthesis and transport
Smooth ER	Tubules, no ribosomes	Lipid and steroid synthesis
Golgi	Stack of flattened sacs + vesicles	Modifies, processes, packages proteins; makes lysosomes
Lysosome	Vesicle of hydrolytic enzymes	Digests worn organelles and engulfed material
Ribosome	80S eukaryote / 70S prokaryote; no membrane	Translation
Centriole	Two microtubule bundles at right angles	Forms the spindle
Chloroplast	Double membrane; thylakoids in grana; stroma	Photosynthesis
Cell wall	Cellulose microfibrils	Support; prevents bursting; freely permeable
Vacuole	Large, bounded by tonoplast	Cell sap; maintains turgor

SHEET 2 — PROKARYOTES & PROTEIN SECRETION (19/21 and 10/21)

Feature	Prokaryotic	Eukaryotic
Nucleus	None ; circular DNA free in cytoplasm, no histones	True nucleus; linear DNA with histones
Organelles	No membrane-bound organelles	Present
Ribosomes	70S	80S
Cell wall	Peptidoglycan (murein)	None in animals; cellulose in plants
Plasmids	Often present	Absent
Size	0.5–5 µm	10–100 µm

PROTEIN SECRETION PATHWAY — 4 MARKS, IN ORDER Nucleus (mRNA out through a nuclear pore) → **ribosome on rough ER** (protein made) → **vesicle** buds off → **Golgi** (**modified**, e.g. carbohydrate added) → **secretory vesicle** → fuses with **cell surface membrane** → **exocytosis**.

A protein-secreting cell has **many mitochondria** (ATP), **extensive rough ER** and a **prominent Golgi**.

SHEET 3 — MICROSCOPY (19/21 papers)

THE TWO DEFINITIONS THEY TEST THE DIFFERENCE BETWEEN **Magnification** = how many times larger the image is than the object.

Resolution = the **minimum distance between two points** at which they can still be distinguished as separate. Limited by the **wavelength** used.

Why EMs resolve better: electrons have a much shorter wavelength than light.

	Light	TEM	SEM
Resolution	~200 nm	~0.1 nm	~3–10 nm
Specimen	Living or dead; colour	Dead, vacuum, very thin section	Dead, surface coated
Image	2D colour	2D black and white	3D surface , black and white
Limitation	Low resolution	No living specimens; artefacts	Lower resolution than TEM

magnification = image size ÷ actual size · 1 mm = 1000 µm = 1 000 000 nm

Measure the image in mm → **convert first** → divide. With a scale bar: measure the bar, divide by what it represents.

SHEET 4 — GAMETES & FERTILISATION (21/21 papers · "acrosome" ×25 in mark schemes)

Sperm	Function	Egg	Function
Acrosome	Hydrolytic enzymes digest the zona pellucida	Zona pellucida	Glycoprotein layer; hardens after fertilisation
Many mitochondria	ATP for the flagellum	Cortical granules	Release contents → thicken zona → prevent polyspermy
Flagellum	Propulsion	Lipid droplets	Energy and materials for the embryo
Haploid nucleus	Restores diploid number on fusion	Follicle cells	Protective outer layer
Small, little cytoplasm	Streamlined, less resistance	Large cell, haploid nucleus	Volume supports early development

FERTILISATION — 5 STEPS **1.** Sperm binds the zona pellucida. **2. Acrosome reaction** — **hydrolytic enzymes** released, **digest through the zona pellucida**. **3.** Sperm and egg **membranes fuse**; nucleus enters. **4. Cortical reaction** — cortical granules release contents, zona **thickens/hardens, preventing polyspermy**. **5.** Haploid nuclei **fuse** → **diploid zygote**.

SHEET 5 — CELL CYCLE & MITOSIS (19/21 · mitotic index in 12/21)

Stage	What happens
G1	Cell grows; organelles replicate; proteins made
S	DNA replicates → each chromosome becomes two sister chromatids
G2	Further growth; preparation for division
Prophase	Chromosomes condense , become visible · nuclear envelope breaks down · centrioles to poles, spindle forms
Metaphase	Chromosomes line up at the equator , attached by centromeres to spindle fibres
Anaphase	Centromeres divide; sister chromatids pulled to opposite poles as fibres shorten
Telophase	Chromosomes decondense; nuclear envelopes re-form
Cytokinesis	Cytoplasm divides → two genetically identical diploid cells

Mitotic index = cells in mitosis ÷ total cells (×100 for %). Count only P, M, A, T — **interphase cells are not in mitosis**. High index = rapid division (root tip, tumour).
Purposes of mitosis: growth · repair and replacement · asexual reproduction.

SHEET 6 — MEIOSIS & VARIATION (9/21 papers, but "crossing over" ×34 in mark schemes)

Two divisions → **four haploid, genetically different** cells from one diploid cell.

THREE SOURCES OF VARIATION — NAME ALL THREE WITH THEIR STAGE **1. Crossing over** — **prophase I**: homologous chromosomes pair as **bivalents, chiasmata** form, **sections of chromatid exchanged**.
2. Independent assortment — **metaphase I**: homologous pairs line up in **random orientation** (and again for chromatids in metaphase II).
3. Random fertilisation — any sperm may fuse with any egg.

	Mitosis	Meiosis
Divisions / cells	One → two	Two → four
Chromosome number	Diploid, unchanged	Halved , diploid → haploid
Genetic identity	Identical	Different
Homologous pairing	None	Prophase I, with crossing over
Purpose	Growth, repair, asexual reproduction	Gamete production

SHEET 7 — STEM CELLS, GENE EXPRESSION & VARIATION (10/21 and 11/21)

Type	Can become	Source
Totipotent	Any cell type, including placenta	Zygote, very early embryo
Pluripotent	Any body cell, not placenta	Inner cell mass of the blastocyst
Multipotent	Limited range of cell types	Adult tissue, e.g. bone marrow

WHY CELLS DIFFER DESPITE IDENTICAL DNA All cells contain the **same genes**, but **different genes are expressed** in each cell type, so **different proteins** are made. Controlled by **transcription factors** binding to a specific DNA region and switching transcription **on or off**. Epigenetically: **methylation** switches off, **histone acetylation** switches on.
Never write "they have different DNA".

Stem cell ethics: FOR — treats currently untreatable conditions; surplus IVF embryos would be destroyed anyway. AGAINST — embryo destroyed; moral status of the embryo; consent; tumour risk. **Then judge**, and note adult/iPS cells avoid much of the objection.

	Continuous	Discontinuous
Control	Polygenic — many genes	Monogenic
Environment	Strong influence	Little or none
Examples / graph	Height, mass, skin colour · normal curve, histogram	Blood group, tongue rolling · bar chart

Phenotype = genotype + environment.

SHEET 8 — BIODIVERSITY (16/21 papers)

- Species richness** = number of different species. Ignores abundance.
- Index of diversity** = accounts for species number **and** individuals of each — the better measure.
- Genetic diversity** = variety of alleles in the gene pool; measured by the **heterozygosity index**.

FORMULA IS GIVEN IN THE PAPER — MARKS ARE FOR USING IT $D = \frac{N(N - 1)}{\sum n(n - 1)}$ · **N** = total individuals of **all** species · **n** = individuals of **each** species
Build an n(n–1) column, sum it, divide. **Higher D = higher diversity**.
Errors: using number of **species** for N · not summing across all species · calculating D but never saying what it means.
Heterozygosity index = heterozygotes ÷ total individuals. Low genetic diversity → less able to adapt to change, more vulnerable to disease.

SHEET 9 — NICHE, ADAPTATION & CLASSIFICATION (14/21 and 11/21)

Niche = the **role and position** of a species in its environment, including all interactions with **biotic and abiotic** factors. Two species cannot occupy the same niche in the same habitat. Not "where it lives".

Adaptation	What it is	Examples
Anatomical	Structural body feature	Thick fur; cactus spines; streamlined body
Physiological	Internal process / biochemical	Concentrated urine; hibernation metabolism
Behavioural	How it acts	Migration; nocturnal activity; basking

Every adaptation answer: feature → how it works → benefit to survival. Naming the feature alone scores nothing.
Hierarchy: Domain · Kingdom · Phylum · Class · Order · Family · Genus · Species. **Three domains:** Bacteria, Archaea, Eukarya — distinguished by **ribosomal RNA sequences**. **Binomial** = *Genus species*.

MOLECULAR PHYLOGENY — THREE METHODS, ONE PRINCIPLE Compare **DNA base sequences** · **amino acid sequences** of proteins (cytochrome c, haemoglobin) · **immunological** comparison.
The more similar the sequences, the more closely related, and the more recent the common ancestor. More reliable than appearance, because similar-looking organisms may have converged.

SHEET 10 — PLANT CELLS, CELLULOSE & FIBRES (19/21 · "cellulose" ×76, "microfibril" ×40 in mark schemes)

Structure	Function
Cell wall	Cellulose microfibrils; support, prevents bursting, freely permeable
Middle lamella	Calcium pectate between adjacent cells, sticking them together
Plasmodesmata	Cytoplasmic channels through the wall — transport and communication
Pits	Thin wall regions allowing water transport
Amyloplast	Stores starch
Vacuole / tonoplast	Cell sap; maintains turgor pressure

WHY CELLULOSE IS STRONG — THE HIGHEST-VALUE PARAGRAPH IN THE UNIT **β-glucose** monomers, **1,4 glycosidic bonds**, alternate molecules **rotated 180°** → chains are **long and straight** → **hydrogen bonds between parallel chains** hold them in **microfibrils** → grouped into **macrofibrils** → very high **tensile strength**.

	Starch	Cellulose
Monomer	α-glucose	β-glucose
Chain shape	Coiled (amylose) / branched (amylopectin)	Long, straight, unbranched
Between chains	Nothing significant	Many hydrogen bonds → microfibrils
Function	Energy storage	Structural — tensile strength

Sclerenchyma fibres — long, thick **lignified** walls, dead at maturity, support. **Xylem vessels** — hollow dead cells, lignified, transport + support.
Plant fibres vs oil-based: **renewable** · **biodegradable** · **less energy** to produce → lower CO₂ · often cheaper. **Against:** variable quality · weaker than some synthetics · land competes with food.

SHEET 11 — MINERALS, DRUG TESTING & ANTIMICROBIALS (10/21, 4/21, 4/21)

Mineral	Needed for	Deficiency
Nitrate	Amino acids and proteins, nucleic acids, chlorophyll	Stunted growth; yellowing of older leaves
Magnesium	Component of chlorophyll	Chlorosis ; reduced photosynthesis
Calcium	Calcium pectate in the middle lamella	Stunted growth; poor roots; distorted young leaves

Trial stage	Tested on	Purpose
Preclinical	Cells, tissues, then animals	Toxicity, basic efficacy
Phase I	Small number of healthy volunteers	Safety, dosage, side effects
Phase II	Small group of patients	Does it work?
Phase III	Large group of patients	Compare with existing treatment or placebo; rarer side effects

DEFINE THESE PRECISELY — "PLACEBO" APPEARED 22× IN THE MARK SCHEMES **Placebo** — a substance with **no active ingredient**, so the drug's effect can be separated from the psychological effect of being treated.
Double-blind — **neither patient nor doctor** knows who got what → **removes bias** from both reporting and assessment.
Randomised — allocation to groups is random → comparable groups, no allocation bias.
Withering: digitalis soup from foxglove, varying proportions to find an effective non-toxic dose. No control, no placebo, no randomisation, no blinding.

Antimicrobial practical: discs soaked in plant extract on inoculated agar; measure the **diameter of the zone of inhibition**; larger = more effective. **Controls:** disc with **solvent only**, and a known antibiotic. **Aseptic:** flame the loop · work near a Bunsen · lift the lid slightly at an angle · tape but don't fully seal · incubate ≤25 °C.

SHEET 12 — CONSERVATION (13/21 papers)

	Methods / conditions	Problems
Seed banks	Seeds dried to low moisture, stored at low temperature (−20 °C) to slow metabolism; viability tested periodically by germinating a sample	Some seeds don't survive drying/freezing · must be periodically regrown, costly · genetic diversity lost over regenerations · doesn't conserve the habitat
Zoos	Captive breeding · studbooks to avoid breeding relatives · IVF and embryo transfer · reintroduction · research and education	Small gene pool → inbreeding · captive-bred animals may not learn natural behaviours · disease spreads · expensive · reintroduction fails if habitat still degraded · ethics of confinement

In situ = in the natural habitat (reserves, national parks) — preserves habitat and natural behaviour, generally preferred. **Ex situ** = outside it (zoos, seed banks, botanic gardens) — a safeguard when in situ isn't feasible.

SHEET 13 — CALCULATIONS (21/21 papers, ~3.8 each)

Type	Method	Trap
Magnification (commonest)	image ÷ actual; measure in mm, convert first	Not converting before dividing
Scale bar	Measure the bar ÷ what it represents = magnification	Sloppy measurement propagates everywhere
Mitotic index (12 papers)	cells in mitosis ÷ total cells	Counting interphase cells
Index of diversity	$N(N-1) \div \sum n(n-1)$	Using number of species for N
Heterozygosity index	heterozygotes ÷ total individuals	Confusing with index of diversity
Volume / surface area	Formula given — substitute carefully	Diameter where radius is wanted
Percentage change	change ÷ original × 100	Dividing by the final value
Cells after n divisions	2 ⁿ	Confusing divisions with cells
Error bars	Overlapping → difference probably not significant	Claiming significance when they overlap

KNOW WHICH TEST DOES WHAT — MULTIPLE-CHOICE ITEMS TEST EXACTLY THIS **Index of diversity** — compares biodiversity between habitats · **Mitotic index** — proportion of cells dividing · **Chi-squared** — observed vs expected · **Student's t-test** — compares means of two independent groups · **Correlation coefficient** — strength of a relationship · **Standard deviation** — spread about the mean

Scientific drawings: sharp pencil · clear **unbroken** lines, no shading · only what you can see · correct proportions · **ruled label lines that don't cross** and touch the structure · title and magnification or scale bar.

SHEET 14 — PHRASE TRIGGERS

If the question is about...	...the phrase they want is
Cellulose strength	β-glucose · alternate molecules rotated 180° · straight chains · hydrogen bonds between chains → microfibrils
Crossing over	Prophase I · bivalents · chiasmata · sections of chromatid exchanged
Independent assortment	Metaphase I · homologous pairs in random orientation
The acrosome	Releases hydrolytic enzymes that digest through the zona pellucida
Preventing extra sperm	Cortical reaction → zona pellucida thickens → prevents polyspermy
Electron microscopes	Electrons have a shorter wavelength → higher resolution
Resolution	Minimum distance between two points still distinguishable as separate
Why cells differ	Same genes , but different genes expressed → different proteins
Transcription factors	Bind a specific DNA region and switch transcription on or off
A diversity value	Higher D = higher diversity ; accounts for species number and abundance
Niche	Role and position in the environment, all biotic and abiotic interactions
Any adaptation	Feature → how it works → benefit to survival
Relatedness from molecules	More similar sequences → more closely related → more recent common ancestor
Placebo / blinding	No active ingredient; neither patient nor doctor knows → removes bias
Anaphase	Centromeres divide; chromatids pulled apart as spindle fibres shorten
Describing a graph	Quote figures with units and name the trend

SHEET 15 — 60-SECOND SELF-TEST

Cover the right column. Hesitate on one, go back to that sheet.

Question	Answer
What does the nucleolus make?	Ribosomes
Function of the Golgi?	Modifies, processes and packages proteins; makes lysosomes
Ribosome size in prokaryotes vs eukaryotes?	70S vs 80S
Prokaryotic cell wall material?	Peptidoglycan (murein)
Order of the secretion pathway?	Nucleus → rough ER ribosome → vesicle → Golgi → secretory vesicle → exocytosis
Define resolution.	Minimum distance between two points still distinguishable as separate
Why do EMs resolve better than light microscopes?	Electrons have a much shorter wavelength
Magnification formula?	Image size ÷ actual size
How many µm in 1 mm?	1000
What does the acrosome contain and do?	Hydrolytic enzymes; digests through the zona pellucida
What prevents polyspermy?	The cortical reaction — zona pellucida thickens
What happens in S phase?	DNA replicates
What divides in anaphase?	The centromeres; sister chromatids separate
Mitotic index formula?	Cells in mitosis ÷ total cells
Three sources of variation in meiosis?	Crossing over (prophase I), independent assortment (metaphase I), random fertilisation
How many cells does meiosis produce, and are they identical?	Four; genetically different
Totipotent vs pluripotent?	Totipotent can form placental tissue; pluripotent cannot
Why do differentiated cells differ?	Same genes, different genes expressed
Index of diversity formula?	$D = \frac{N(N-1)}{\sum n(n-1)}$
What is N in that formula?	Total number of individuals of all species
Define a niche.	The role and position of a species in its environment, with all biotic and abiotic interactions
Three types of adaptation?	Anatomical, physiological, behavioural
The three domains?	Bacteria, Archaea, Eukarya
Which glucose is cellulose made of?	β-glucose
What holds cellulose chains together?	Hydrogen bonds between parallel chains, forming microfibrils
What is the middle lamella made of?	Calcium pectate
Magnesium's role and deficiency symptom?	Component of chlorophyll; chlorosis
Who is tested in Phase I?	A small number of healthy volunteers
Define double-blind.	Neither patient nor doctor knows who received the drug
Two seed bank storage conditions?	Dried to low moisture; stored at low temperature (−20 °C)
What is a studbook for?	Recording ancestry so related individuals are not bred together
Which test compares observed with expected?	Chi-squared

WALKING INTO THE EXAM Underline the command word. Count the marks, then count your points. Convert units before every magnification calculation. Give a reason wherever "explain" appears, and state the benefit to survival in every adaptation answer. Leave Q1-Q3 behind on time — Q5-Q8 hold about 60% of the paper.